

Designing Cognitive Mini-Games for Computational Thinking and Executive Functions (Work in Progress)

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Abstract

Computational Thinking (CT) has been defined as one of the key competences for successfully navigating the 21st century, as it holds strong potential as a transversal skill and offers promising benefits for the development of key cognitive abilities. Despite these advantages, the integration of CT into current curricula remains fragmented, lacking a holistic approach that combines both the technical and cognitive dimensions of CT. Existing tools tend to focus on the technical aspects, using plugged and unplugged activities to address specific CT components or content (e.g., functions or algorithms), while often neglecting the underlying cognitive mechanisms, such as executive functions (EFs). To address this gap, we have been working on the ATHEnA project, which stands for *Algorithmic Thinking, cognitive Enhancement and Active learning*, a game-based experience that integrates CT concepts with EF training. The project consists of a set of minigames embedded within a narrative framework, designed to support children aged 6–12 in developing both the concepts and processes underlying CT, while simultaneously enhancing their EFs. In this work-in-progress, these objectives are achieved through participatory design (with students, teachers and experts) of the game mechanics and narratives, which are grounded in validated literature on gamified cognitive training and evidence-based approaches to CT education.

CCS Concepts

• **Applied computing** → **Interactive learning environments**.

Keywords

Serious Games, Game-based learning, Computational Thinking, Executive Functions.

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1 Introduction

The pervasive influence of technology has called for a redefinition of the skills that educational systems need to equip their students

with. In the context of curriculum development, *computational thinking* (CT) emerges as transversal competences to be developed, since it goes beyond the instrumental use of computers, referring to the cognitive processes involved in formulating and evaluating step-by-step solutions. The majority of tools for CT development focus on code-driven tasks and tend to neglect the cognitive dimension. To address this gap, we developed the idea of ATHEnA, which stands for *Algorithmic Thinking, cognitive Enhancement and Active learning*. While CT is often taught through programming, CT development also involves domain-general cognitive resources, such as working memory, inhibitory control, and cognitive flexibility, that support planning, monitoring, and revising problem-solving strategies. ATHEnA is designed around this premise: it treats executive functions (EFs) not as ancillary outcomes, but as *design targets* that shape how CT concepts are practiced in play. Our contribution is therefore not simply ‘mini-games for learning,’ but a mechanism-driven design approach that links CT goals to EF constraints and operationalizes this mapping in concrete game mechanics. The proposal includes a set of five minigames designed to foster cognitive development. Through scaffolded and engaging gameplay, the suite introduces children to core CT concept, such as decomposition, pattern recognition, abstraction, algorithms, debugging, conditionals, and loops, while simultaneously supporting the growth of *executive functions* (EFS).

Contribution. This paper presents the work-in-progress research and design behind the ATHEnA project, aimed at providing a cognitive informed toolkit for CT education. This contribution focuses on the initial phase of the project including the design rationale and framework as well as the initial participatory design data guiding the upcoming phases. We report the design rationale, the CT-EF-to-mechanics mapping that structures the intervention, and early participatory-design insights that have already shaped the narrative and interaction patterns.

- *A systematic mapping between CT concepts, executive functions, and game mechanics*, illustrating how high-level cognitive constructs are translated into concrete, observable player actions through specific mechanics.
- *A set of mini-game design archetypes* that practice CT concepts through constrained, feedback-rich interactions that reduce reliance on programming syntax and enable targeted training of planning, monitoring, and strategy shifting.
- *A participatory design-informed development process*, currently underway, involving primary school children and teachers to refine mechanics and narrative elements, ensuring engagement and pedagogical coherence.



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2 Theoretical Background

CT refers to the cognitive process of formulating solutions to problems in the form of precise, step-by-step procedures intended to be executed by an agent, whether human or machine [3]. CT has five underlying processes: decomposition, abstraction, pattern recognition, algorithm design, and debugging which can be transferable and supporting to problem solving across different domains [5]. Furthermore, CT involves a set of practices, including simulation, evaluation, experimentation, iteration, tinkering, and especially debugging [12], which offers a natural opportunity for learners to reflect, revise, and improve their strategies.

2.1 Game-Based Learning and Tools for CT Education

Existing tools used to teach CT often present structural and pedagogical limitations. Many are code-based and require slow, reflective execution, or they offer blank-canvas environments, such as Scratch or educational robotics kits, which, while flexible, are often too open-ended and time-consuming to effectively target specific CT components in short- or medium-length learning contexts. Most widely used CT tools follow a common approach:

- Code.org (<https://code.org/>) includes step-based tutorials and puzzles focused on syntactical practice but limiting broader CT development;
- Scratch (<http://scratch.mit.edu/> [12]) offers a block-based environment fostering creativity but heavily relying on student initiative and prior structure;
- Bebras tasks (<https://bebras.it/>) provide problem-solving exercises focused on logic and abstraction but detached from procedural or iterative CT practices;
- CS Unplugged (<https://www.csunplugged.org/en/>) activities use physical materials to illustrate CT concepts without computers, but sometimes lacks depth and feedback mechanisms to support metacognitive reflection.

These tools have in common to emphasize either solving tasks or creating from scratch, with few directly addressing core CT concepts or making the cognitive processes behind problem-solving visible. None of them explicitly aim to support the metacognitive dimension CT, which is increasingly recognized as essential for learning transfer and deeper understanding. This gap sheds light on the need for new educational experiences that embed CT concepts into more scaffolded, cognitively grounded, game-based activities.

2.2 Cognitive Aspects of Computational Thinking

The exploration of the cognitive underpinnings of programming and their connection to the theoretical foundations of problem solving began in the late 1970s. Although programming and CT are often discussed together, they need to be distinguished: CT refers to the underlying mental processes, while programming is one of the contexts in which these processes are applied. These domains have increasingly been investigated by psychology and cognitive science, shedding light on how programming, and especially CT, can enhance cognitive skills such as reasoning, logical thinking, planning, and problem solving [9], as well as broader higher-order

thinking skills [7]. CT has also been shown to influence creative thinking and metacognition [10]. However, a precise definition of the cognitive underpinnings of CT remains challenging, given its nature as a higher-order thinking skill [5]. The connection between CT and EFs appears to be particularly important, as working memory, cognitive flexibility, and inhibitory control are essential for successfully carrying out these problem-solving strategies [4]. Teaching CT through structured and applied approaches may improve key EFs in elementary school children [8]. In the same vein, efforts to investigate the effects of CT activities on EF development have yielded promising results [1, 3].

3 Methodology

The ATHEnA project is articulated along different phases and currently ongoing. The development process includes an initial phase focused on translating executive function-based cognitive mechanisms into computational thinking constructs, grounded in validated literature and evidence-based educational practices [8]. This initial desk work is followed by a series of participatory design workshops involving children, teachers, and domain experts, aimed at co-constructing narratives [19], mechanics, and interaction details to anchor the experience in authentic user perspectives [14]. Lastly, the insights gathered from the previous phases will be informing the implementation and prototype to be evaluated in an empirical setting. At the time of writing, the project is in the participatory design phase, with workshops currently underway.

Participatory Design Workshops with different stakeholders. A series of participatory design workshops are being conducted with the different actors and future users of the ATHEnA project such as elementary school students, teachers and educators and field experts. During the sessions, participants are engaged with low-fidelity materials and produced a range of design artifacts, including drawings and sketches of characters and game worlds, annotated paper prototypes of mini-games, storyboard-based narratives, and mixed artifacts combining visual elements with verbal explanations [19] [18]. The resulting artifacts of the sessions, as well as observational notes and verbal elements have been analyzed qualitatively to extract design-relevant insights concerning engagement, cognitive accessibility, narrative interpretation, and perceived game mechanics [19]. Workshops with teachers are currently conducted.

4 Preliminary Outcomes

4.1 Theoretical rationale: From Computational Thinking and Executive Functions to Game Mechanics

A pivotal challenge in the ATHEnA project lays in translating abstract cognitive constructs into concrete, playable game mechanics aligned with theoretical models and empirical evidence. The operative relationships between CT and EFs is still not well defined, however, guidance on this matter was found by adopting a mechanism-driven design approach, in which each mini-game mechanic is intentionally derived from a specific cognitive demand [17]. Building on this, in ATHEnA mini-games CT components define the epistemic goal of the task (e.g., decomposing a problem,

identifying patterns, planning an algorithm), while executive functions determine the cognitive constraints under which that task is performed (e.g., maintaining information in working memory, inhibiting prepotent responses, shifting strategies based on feedback) [17]. Game mechanics are then designed to enforce these constraints through time pressure, rule changes, limited action windows [6] [15]. For instance, algorithmic thinking is operationalized not merely by asking players to produce a correct sequence of actions, but by requiring them to plan, store, and execute that sequence under working memory constraints and on the same line debugging is enacted through requests to the player of monitoring system states, and flexibly revise strategies by engaging cognitive flexibility and inhibitory control. On a higher level, The ATHEnA project is rooted in constructionist learning theory and its game mechanics are built upon scaffolding principles [15]. More in detail, the design and mini-games' mechanics are informed by literature on cognitive training and action-video games (AVG) employed for enhancing EFs [2, 6, 16, 17]. Furthermore, the ATHEnA project's tasks draw inspiration from effective interventions for CT education focusing on the type of tasks and requirements of inclusive, and play-based pedagogies. In fact, while tangible coding tools and robotics have shown benefits in CT education [3], they often depend on adult mediation and external guidance. ATHEnA, by contrast, is designed to be intuitively usable by children themselves, supporting autonomy and exploration through a narrative-based game world. In addition, ATHEnA critically addresses a gap of many CT platforms which commonly emphasize either solving predefined tasks or providing open-ended coding environments which beyond programming syntax and logic, they do not integrate the cognitive underpinnings. Therefore, ATHEnA's structure bridge this gap by embedding CT concepts directly into cognitively meaningful activities, allowing children to engage with the core elements of CT while simultaneously strengthening EFs. The ATHEnA project addresses the educational gaps in non-code-centered CT and EF training, striving to offer a gamified experience through which players gradually become familiar with and automate CT concepts and processes such as decomposition, pattern recognition, abstraction, algorithms, debugging, variables, conditionals, events, and loops, while simultaneously training EFs. In this context, ATHEnA is intended to complement code-based approaches with a structured for practicing CT strategies alongside programming activities. These principles are then implemented in a set of five standalone, short minigames.

4.1.1 Game Structure and Mechanics. Drawing upon the previous paragraph, ATHEnA presents a set of five minigames in addition to a cognitive activation task presented at the beginning of each session [16] [15]. Each mini-game is set in a planet and poses a different challenge doable only through the application of CT strategies. While games can be played independently, they are embedded in a connected narrative universe promoting the exploration of the different processes as well as concepts scaffolding. Mini-games mechanics and features are presented in 1.

4.1.2 Participatory Design Insights. The artifacts and qualitative insights collected during the participatory design workshops conducted until the moment of writing provided key elements for the narrative structure of ATHEnA. The insights brought by children



Figure 1: Preliminary ATHEnA mini-games and narrative overview

depicted narratives usable as a cognitive scaffold to interpret sequencing, causality, rule changes, and error correction, suggesting that tightly coupling story and mechanics can support both computational thinking and executive function engagement. The animal detective in a space setting narrative emerged organically from participatory design and was retained because it provides an intuitive, emotionally safe metaphor for investigation, rule discovery, and debugging, core practices of computational thinking. By making the link between these mechanics and CT explicit, ATHEnA ensures that narrative elements function as cognitive scaffolds supporting computational and executive processes.

Preliminary Design Choices. The overall narrative and theme of ATHEnA revolve around a space adventure set in a distant universe, home to the *Computational Galaxy*: a constellation of small planets, each embodying specific concepts and processes of CT (Fig. 1). This galaxy is inhabited by unique, colorful, geometrically shaped creatures called *Bitlings*, whose everyday activities reflect computational processes. The insights collected with elementary school children guided the refinement of this narrative framework by reinforcing a storyline centered on small agents collaboratively protecting meaningful places through exploration and reasoning.

5 Discussion and Limitations

ATHEnA aims at filling a critical educational gap at the intersection of CT and EFs by shifting the focus from code-based exercises to transversal cognitive empowerment by leveraging intuitive gameplay to train both CT components and EFs. The adopted approach offers a structured, low-barrier entry point for young learners to engage with abstract concepts in a playful and embodied way by grounding each mini-game in validated cognitive mechanics. The modular format of mini-games supports diverse deployment scenarios: from in-class use to informal learning settings, after-school programs, or even individual home play [15] [16]. Immediate feedback mechanisms and adaptive difficulty levels to properly challenge learners and maintain engagement, while allowing educators to integrate the tool without requiring specialized training. The insights gained from the workshops conducted first with elementary school students, guided the direction of the narrative structure, which centers on small investigative characters collaboratively protecting their space environment [18]. Nonetheless, the project faces a set of design tensions. First, the balance between cognitive challenge and game enjoyment and second achieving generalizability of CT concepts while supporting conceptual clarity and transfer.

Mini-Game	CT Concept	EFs Engaged	Game Mechanics
Mission Warm-Up	–	Cognitive and attentional activation	At the beginning of each game session, the player receives signals from the mysterious galaxy and must “decode” a message sent by the Bitlings. Signals can be colors, shapes, or numbers, and the player must monitor the sequence and identify when the current signal matches the one that appeared n steps earlier (e.g., 1-back, 2-back).
Break It Down	Decomposition	Planning, sequential and sustained attention	Given a complex shape, the player must select all the elements it is composed of in order to make them pass through a door. Across levels, difficulty increases in terms of the number of elements and perceptual ambiguity: elements initially differ in color and outline, but gradually become uniform in color and lack visual boundaries.
Glitchers Attack	Pattern recognition, abstraction, events, debugging	Monitoring, inhibition, cognitive shifting, and working memory updating	Inspired by the Wisconsin Card Sorting Test (WCST), this mini-game requires players to rapidly categorize incoming “Glitchers” according to hidden rules that periodically change (e.g., color, shape, or number). Each Glitcher carries a symbol, and the player must infer and select the correct matching rule from a set of options. Correct rule switching after feedback triggers a “pattern recognition burst” that eliminates visible enemies, while errors cause enemies to advance, increasing pressure and reinforcing learning through feedback.
Plan It Path	Algorithms	Working memory and planning	The Bitlings are trapped in a maze while Glitchers approach. The player must plan a complete sequence of commands (e.g., move, turn, jump, activate) in advance to guide them to safety before execution, without the possibility of step-by-step correction.
Rocket Run	Debugging, variables, and conditionals	Monitoring and cognitive flexibility	The player acts as a spaceship operator responsible for maintaining optimal system functioning via a control panel under constantly changing conditions. Unpredictable system alerts require monitoring errors and identifying malfunctioning components through a debugging process involving variable inspection, if-then logic (conditionals), and the selection of appropriate action sequences.
Loop Escape	Pattern recognition and loops	Cognitive flexibility and sustained attention	Glitchers move according to a predictable, looping pattern over time. The player must observe and identify this pattern to move Bitlings into shelters during safe time windows. After two full iterations, the pattern changes, requiring players to adapt their strategy.

Table 1: Overview of ATHEnA project mini-games, including targeted Computational Thinking concepts, executive functions engaged, and underlying game mechanics.

6 Conclusions and Future Work

This contribution explored the design rationale of ATHEnA proposing a model where CT becomes not only a set of technical skills but also a pathway to enhance EFs. The proposal responds also to the practical limitations of many current CT educational tools, positioning the project as a promising step toward inclusive, flexible, and cognitively grounded learning tool for CT. Future work will move toward three main directions: the completion and release of a fully functional demo, the launch of empirical validation studies

and the beginning with pilot testing to assess the educational impact on CT, EF development across time. In this regard, the final version of the ATHEnA game will undergo scientific validation involving classroom-based testing with a larger group of students to assess usability, engagement, and educational effectiveness. In this sense, ATHEnA is envisioned not merely as a game, but as a scalable educational framework capable of fostering cognitive and computational literacy from the earliest stages of learning.

References

- [1] B. Arfé, T. Vardanega, C. Montuori, and M. Lavanga, “Coding in primary grades boosts children’s executive functions,” *Front. Psychol.*, vol. 10, p. 2713, 2019.
- [2] S. Bertoni, S. Franceschini, G. Puccio, M. Mancarella, S. Gori, and A. Facchetti, “Action video games enhance attentional control and phonological decoding in children with developmental dyslexia,” *Brain Sci.*, vol. 11, no. 2, p. 171, 2021.
- [3] M. C. Di Lieto et al., “Empowering executive functions in 5- and 6-year-old typically developing children through educational robotics: An RCT study,” *Front. Psychol.*, vol. 10, p. 3084, 2020.
- [4] M. Lodi, *Introducing computational thinking in K-12 education: Historical, epistemological, pedagogical, cognitive and affective aspects*. Bologna: Alma Mater Studiorum – Università di Bologna, 2020.
- [5] C. Montuori, F. Gambarota, G. Altoé, and B. Arfé, “The cognitive effects of computational thinking: A systematic review and meta-analytic study,” *Comput. Educ.*, vol. 210, p. 104961, 2024.
- [6] A. Pasqualotto, J. Parong, C. S. Green, and D. Bavelier, “Video game design for learning to learn,” *Int. J. Hum. Comput. Interact.*, pp. 1–18, 2022.
- [7] R. Rianti, Z. A. Aziz, and M. Aulia, “Incorporating higher order thinking skills into English summative assessments,” *Engl. Rev. J. Engl. Educ.*, vol. 12, no. 1, p. 1, 2024.
- [8] C. Robledo-Castro, C. Hederich-Martínez, and L. F. Castillo-Ossa, “Cognitive stimulation of executive functions through computational thinking,” *J. Exp. Child Psychol.*, vol. 235, p. 105738, 2023.
- [9] K. Tsarava et al., “Cognitive correlates of computational thinking: Evaluation of a blended unplugged/plugged-in course,” in *Proc. 14th Workshop Prim. Second. Comput. Educ.*, 2019, pp. 1–9.
- [10] A. Yadav, C. Ocak, and A. Oliver, “Computational thinking and metacognition,” *TechTrends*, vol. 66, no. 3, pp. 405–411, 2022.
- [11] M. Zapata-Caceres, E. Martin-Barroso, and M. Roman-Gonzalez, “Collaborative game-based environment and assessment tool for learning computational thinking in primary school: A case study,” *IEEE Trans. Learn. Technol.*, vol. 14, no. 5, pp. 576–589, 2021.
- [12] Resnick, M., Maloney, J., Monroy-Hernández, A., Rusk, N., Eastmond, E., Brennan, K., Millner, A., Rosenbaum, E., Silver, J., Silverman, B., & Kafai, Y. (2009). Scratch: Programming for all. *Communications of the ACM*, 52(11), Artículo 11.
- [13] Román-González, M., Pérez-González, J.-C., & Jiménez-Fernández, C. (2017). Which cognitive abilities underlie computational thinking? Criterion validity of the Computational Thinking Test. *Computers in Human Behavior*, 72, 678–691. <https://doi.org/10.1016/j.chb.2016.08.047>
- [14] Menestrina, Z.; Pasqualotto, A.; Siesser, A.; Venuti, P.; De Angeli, A. Engaging Children in Story Co-Creation for Effective Serious Games. *Sustainability* 2021, 13, 10334. <https://doi.org/10.3390/su131810334>
- [15] Boyle, E.A.; Hainey, T.; Connolly, T.M.; Gray, G.; Earp, J.; Ott, M.; Lim, T.; Ninaus, M.; Ribeiro, C.; Pereira, J.M. An update to the systematic literature review of empirical evidence of the impacts and outcomes of computer games and serious games. *Comput. Educ.* 2016, 94, 178–192.
- [16] Breuer, J.S.; Bente, G. Why so serious? On the relation of serious games and learning. *J. Comput. Game Cult.* 2010, 4, 7–24.
- [17] Anguera, A.J.; Gazzaley, A. Video games, cognitive exercises, and the enhancement of cognitive abilities. *Curr. Opin. Behav. Sci.* 2015, 4, 160–165.
- [18] Van Mechelen, M.; Zaman, B.; Laenen, A.; Abeele, V.V. Challenging group dynamics in participatory design with children: Lessons from social interdependence theory. In *Proceedings of the 14th International Conference on Interaction Design and Children*, Boston, MA, USA, 21–24 June 2015; pp. 219–228.
- [19] Lebowitz, J.; Klug, C. *Interactive Storytelling for Video Games: A Player-Centered Approach to Creating Memorable Characters and Stories*; Taylor & Francis: Abingdon, UK, 2011.